

18ECP109L-PROJECT

RETINAL DISEASE DETECTION AND CLASSIFICATION USING CONVOLUTION NEURAL NETWORK WITH TRANSFER LEARNING

A PROJECT REPORT

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Of

BACHELOR OF TECHNOLOGY

in

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BONAFIDE CERTIFICATE

Certified that this project report titled “**DETECTION AND CLASSIFICATION OF RETINAL DISEASE USING CONVOLUTION NEURAL NETWORK WITH TRANSFER LEARNING**” is the bonafide work of “**ADITYA AGARWAL [Reg No: RA2111004010177], ABHINAV PANWAR[Reg No:RA2111004010210], ROHAN GUPTA [Reg No: RA2111043010033]**, who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION

We hereby declare that the Major Project entitled “**DETECTION AND CLASSIFICATION OF RETINAL DISEASE USING CONVOLUTION NEURAL NETWORK WITH TRANSFER LEARNING**” to be submitted for the Degree of Bachelor of Technology is our original work as a team and the dissertation has not formed the basis of any degree, diploma, associateship or fellowship of similar other titles. It has not been submitted to any other University or institution for the award of any degree or diploma.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iv
	LIST OF TABLES	vi
	LIST OF FIGURES	vii
	LIST OF ABBREVIATIONS	viii
1	INTRODUCTION	1
1.1	BACKGROUND	1
2	LITERATURE REVIEW	3
2.1	OVERVIEW OF OCT IMAGING	3
2.2	EXISTING METHODS IN IMAGE CLASSIFICATION	4
2.3	COMPARATIVE STUDIES	5
3	RESEARCH METHODOLOGY	6
3.1	PROBLEM STATEMENT	6
3.2	SCOPE FOR STUDY	6
3.3	OBJECTIVE OF THE STUDY	7
3.4	REALISTIC CONSTRAINTS	7
3.5	ENGINEERING STANDARDS	7
4	DESIGN AND METHODOLOGY	8
4.1	THEORETICAL ANALYSIS	8
4.2	DATA COLLECTION	9
4.3	MODEL SELECTION	10
4.3.1	CNN	10
4.3.2	GoogLeNet	11
4.3.3	DenseNet	11

4.3.4	ResNet50	12
4.4	SYSTEM ARCHITECHURE	12
4.5	TRAINING, TESTING AND VALIDATION	13
4.6	EVALUTION METRICS	14
5	RESULTS AND DISCUSSION	17
5.1	EXPERIMENTAL RESULTS	17
5.2	MODEL PERFORMANCE	17
5.2.1	Accuracy	18
5.2.2	Comparison with other predictive models	18
5.3	ANALYSIS OF RESULTS	22
5.4	SUGGESTIONS AND RECOMMENDATIONS	27
6	CONCLUSION AND FUTURE WORK	29
6.1	SUMMARY	29
6.2	CONCLUSIONS	30
6.3	FUTURE ENHANCMENTS	30
	REFERENCES	31
	PUBLICATION DETAILS	33

ABSTRACT

Eye diseases have been a significant health concern throughout history, with conditions such as Age-related Macular Degeneration (AMD) increasingly recognized as main causes of the vision impairment worldwide. Traditionally, the assessment of retinal conditions relied heavily on manual evaluations of Optical Coherence Tomography (OCT) images, which, while effective, are often time-consuming and require specialized expertise.

The Categorizing of retinal diseases utilizing OCT images has garnered considerable interest in the domain of computational medical imaging. This project introduces a deep learning methodology for the automated classification of OCT images via a custom Convolution Neural Network (CNN) in conjunction with recognized transfer learning models. Deep learning methodologies are favored for their exceptional accuracy and performance; however, they frequently necessitate substantial computational resources and time for training. We employ MobileNet, ResNet50, GoogleNet, and DenseNet for feature extraction and comparison analysis.

The models are trained and refined on an OCT dataset utilizing various hyperparameter settings, such as learning rate, epoch count and optimizing techniques to get maximal accuracy. Upon concluding our study, we evaluate the performance of these models and ascertain the optimal design based on accuracy and additional assessment criteria. Our methodology illustrates the capability of deep learning in the automated classification of OCT images, providing enhanced diagnostic support for the identification of retinal diseases.

One of the strongest components of this project is the comparison of multiple model classification performance metrics evaluation, including accuracy, precision, recall, F1-score, and confusion matrices. By comparing different architectures in a structured way, we determine the best performing and accurate deep learning model for retinal detection of OCT images. The outcome of this research contributes to the literature in AI-assisted ophthalmology, offering a cost-efficient and scalable solution for early detection and screening of retinal disease. This project supports the Sustainable Development Goal (SDG) 3: Good Health and Well-being goal by promoting the utilization of new and advanced technology in healthcare in order to facilitate timely diagnosis, support patient outcomes, and improve the quality of healthcare services and practice. Additionally, with the automation of deep learning AMD detection, this project can support reducing preventable blindness through early detection of AMD and contribute to sustainable health practices.

LIST OF FIGURES

Figure 2.1 Abnormal eye.....	3
Figure 2.2 Normal eye.....	3
Figure 4.1 CNN Architecture.....	10
Figure 4.2 GoogleNet Architecture.....	11
Figure 4.3 DenseNet Architecture.....	11
Figure 4.4 ResNet Architecture.....	12
Figure 4.6 Project Workflow.....	13
Figure 5.1 Model Comparison Graph.....	21
Figure 5.2 Model Training and Testing Accuracy.....	21
Figure 5.3 Model Training and Testing Loss.....	22
Figure 5.4 Confusion Matrix	24
Figure 5.5 ROC Curve	24
Figure 5.6 Accuracy Curve.....	25
Figure 5.7 Loss Curve	25
Figure 5.8 Web Application Desktop Version	27
Figure 5.9 Web Application Output Image.....	27

LIST OF TABLES

Table 4.1 Training and Testing Model Metrics	14
Table 5.1 Comparison of DenseNet Model	19
Table 5.2 Comparison of CNN Model.....	20
Table 5.3 Comparison of ResNet50 Model	20
Table 5.4 Comparison of GoogleNet Model	20

ABBREVIATIONS

AMD - Age-related Macular Degeneration

OCT – Optical Coherence Tomography

CNN- Convolution Neural Network

ResNet50 – Residual Network

ROC/AUC - Receiver Operating Characteristic/Area Under Curve

DME – Diabetic Macular Edema

CNV – Choroidal neovascularization

RNN – Recurrent Neural Network

CHAPTER 1

INTRODUCTION

1.1 Background

The development of medical imaging has revolutionized diagnostics, particularly in ophthalmology. Initially, retinal conditions were diagnosed through direct observation, which was limited by the quality and depth of the physician's view. The introduction of OCT in the early 1990s marked a significant advancement, enabling high-resolution cross-sectional imaging of retinal layers. This innovation provided unprecedented detail, allowing clinicians to detect and monitor diseases like age-related macular degeneration (AMD) with greater accuracy.

Eye diseases have been a severe public health issue for decades, affecting millions of individuals globally. Of these, Age-related macular degeneration (AMD) has been one of the main sources of vision loss and blindness, primarily among the elderly. Although these methods tell that all the expert assessments are extremely specific, consuming, labor-intensive, and prone to inter-observer variability, leading to preventable delay in diagnosis and therapy.

With the rapid advancement of artificial intelligence (AI) and deep learning, computational approaches have revolutionized the field of medical imaging to offer computerized solutions for disease detection and classification. Convolutional Neural Networks (CNNs) have proved phenomenal in handling intricate medical image data, feature hierarchies, and high-precision classification. Deep learning approaches have gained widespread use in the detection of retinal disease, where they outperform traditional machine learning models significantly by reducing handcrafted feature extraction and enabling end-to-end learning from raw images.

This work introduces a deep learning framework for the automatic OCT image classification, using a customized CNN model and well-known transfer learning models. Transfer learning uses pre-trained models to effectively learn features